Palindromic length in free groups, Catalan expressions, and new risks for mathematicians

Volker Diekert, **Anna Frid**, Silas Natterer and Alexander Thumm

September 29, 2026

Palindromic length

The palindromic length $pl(u)$ is the minimal number of concatenated palindromes needed to express the word u.

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Do we consider a free semigroup or a free group? Or maybe something else?

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The difference can be unbounded [Saarela, 17]:

$$AbcA^{\rho} = (a_1 \cdots a_n)bc(a_n \cdots a_1) = (AbA^{\rho})(A^{\iota\rho}A^{\iota})(AcA^{\rho}).$$

$2n + 2$ vs. 3 palindromes

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- can be computed in linear time [Borozdin, Kosolobov, Rubinchik, Shur, 2017];
- is unbounded for factors of aperiodic infinite words [Rukavicka, 2026].

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- unbounded on $aba^2b^2 \cdots a^n b^n$;
- [Saarela 2017]: important examples.

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- Around 2018: I learn about the question in a private communication with Valery Bardakov.
- 2019: a characterization of words of palindromic length 2 and a conjecture for the pal. length 3 are given in a Master thesis of my student M. Bou-Laouz.

First preprint

math.CO] 10 Dec 2025

Small palindromic lengths in free groups and word equations with antimorphisms

Anna E. Frid*

December 12, 2025

Abstract

The palindromic length of a finite word w is defined as the minimal number of palindromes such that their product is w. Clearly, this function may take different values depending on if we consider w as an element a free semigroup or of a free group: for example, in the free semigroup, the palindromic length of $abca$ is 4 (here every letter is a palindrome), and in the free group, it is 3 since $abca = (aba)(a^{-1}a^{-1})(aca)$.

In free semigroups, the palindromic length can clearly be computed, and there are fast algorithms for that. In free groups, the question is trickier. In this paper, we characterize words in the free group whose palindromic length is 2 and 3.

Conjecture deep inside

So, with possible forms of words of palindromic lengths 2 and 3, and with a particular conjecture on the palindromic length 4, I also stated the following:

> This fact and the first values $1, 1, 2$ and perhaps 5 of the number of possible expressions for $k = 1, 2, 3, 4$, suggest that perhaps the reduced forms may somehow be related to Dick paths, and their numbers are Catalan numbers.

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- September 11: I announce this talk online:

Upcoming talks | Past talks

Your time	Speaker	Title	Saved
Tue Sep 29 15:00	Anna Frid	Palindromic length in free groups and Catalan expressions	
Tue Oct 13 15:00	Vanessa Kosoy	TBA	

Surprise!

15 Sep 2026

Palindromic Length in Free Groups: Reflections, Noncrossing Matchings, and Catalan Forms

Junjie Liao
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Abstract

Let $F = F(X)$ be a free group of finite rank, with palindromic length taken with respect to the fixed basis X. We embed F as the index-two subgroup of the universal Coxeter group $W = F \rtimes_\theta \langle t \mid t^2 = 1 \rangle$, where $\theta(x) = x^{-1}$ for $x \in X$, and prove

$$\mathrm{pl}(g) = \min\{\ell_T(g), \ell_T(gt)\}.$$

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- But it seems that even announcing titles can be risky in the age of AI (if we consider it as a risk).

Initial result

Theorem 1

A word is a concatenation of 2 palindromes in the free group if and only if its reduced form looks like

$$xp_1p_2x^\iota$$

for some (possibly empty) palindromes p_1, p_2 and a (possibly empty) word x.

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$\Longrightarrow$: $xp_1p_2x^\iota = (xp_1x^\rho)\ (x^{\iota\rho}p_2x^\iota)$.

Proof (one of the cases)

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$\boxed{xy^\rho = x p_1^\iota p_2^\iota x^\iota}$

Three palindromes

Theorem 2 [F., 2025]

A word is a concatenation of 3 palindromes in the free group if and only if its reduced form looks like

$$x_1 x_2 p_1 p_2 x_2^{\iota} p_3 x_1^{\rho}$$

or

$$x_1 p_1 x_2 p_2 p_3 x_2^{\iota} x_1^{\rho}$$

for some (possibly empty) palindromes p_1, p_2, p_3 and a (possibly empty) words x_1, x_2.

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The proof is technical with many cases and subcases.

Catalan expressions

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A P-expression is a *Catalan* one if it is minimal and contains a maximal number of parentheses:

$$((pp)pp)p \mapsto (((pp)(pp))p) \text{ or } ((((pp)p)p)p).$$

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  and $(((pp)p)p)$, $((pp)(pp))$, $((p(pp))p)$, $(p(p(pp)))$, $(p((pp)p))$ for $n = 3$.

Decompositions

Let u be a P-expression.
A *u-decomposition* of a word $w \in A^*$ is a decomposition of w to reduced factors according to the letters of u as follows:

- every factor corresponding a letter p is a palindrome;
- the factors corresponding to a pair of parentheses are of the form x and x^α, where

$$\alpha = \begin{cases} \iota, & \text{if the number of palindromes inside is even,} \\ \rho, & \text{if it is odd.} \end{cases}$$

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$$(((pp)p)p) \mapsto x_1x_2x_3p_1p_2x_3^\iota p_3x_2^\rho p_4x_1^\iota.$$

General result

Theorem

A word is a concatenation of at most n palindromes in the free group if and only if its reduced form admits a u-decomposition for a P-expression u with n letters p.

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$$ababbbabbabbab^{\iota}aab^{\iota}abbba^{\iota} = a \cdot b \cdot \cdot abbba \cdot bbabb \cdot \cdot ab^{\iota}aab^{\iota}a \cdot b \cdot bb \cdot a^{\iota}$$

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4. $(E_1)(E_2), ((E_1)E_2), (E_1(E_2)) \to ((E_1)(E_2))$.

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Lemma

Let $w \in A^*$ admit a u-decomposition to a P-expression u. Then either w or a reduction of w admits a decomposition to a minimal P-expression with the same number of palindromes.

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So, the theorem holds.

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There are $O(n^2)$ such decompositions.

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We have $pl(w) \leq k$ if and only if there exist u and v with $pl(u) = m$ and $pl(v) = k - m$ for some m (with the correct correct α).

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+ dynamic programming to compute pl of each factor only once. □

Conclusion

- So, we found a beautiful combinatorial description of words with free group palindromic length $\leq n$: they have the structure of a Catalan expression with $n-1$ pairs of parentheses.

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- My coathors have also other results, more sophisticated and general.
- But in the era of AI, we cannot announce them or even give keywords before the preprint is online.